

Truth Functions

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Let us write $\alpha(p_1, \dots, p_n)$ to denote that the wff α has the propositional variables $p_1, \dots, p_n$.

Definition 0.1. A *truth function* is a map $f : \{T, F\}^n \longrightarrow \{T, F\}$, where $n \in \mathcal{N}$, the set of natural numbers.

Example: $n = 2$. $f(t_1, t_2) := t_1 \wedge t_2$, where $\wedge$ is defined by the truth table given earlier.

Notation. Let f_* denote the truth function corresponding to a logical connective $*$. (So f in the above example would be denoted by $f_\wedge$.)

Exercise 0.1. Given n , what is the total number of truth functions from $\{T, F\}^n$ to $\{T, F\}$?

We now show that for each wff α , there is a truth function f_α , and, on the other hand, for every truth function f , there is a wff α such that $f_\alpha = f$.

Proposition 0.1. Every wff $\alpha(p_1, p_2, \dots, p_n)$ gives rise to a unique truth function $f_\alpha : \{T, F\}^n \longrightarrow \{T, F\}$ that represents the truth table of α .

Proof. If $(t_1, t_2, \dots, t_n) \in \{T, F\}^n$, define a valuation $v : PV \longrightarrow \{T, F\}$ such that $v(p_i) := t_i$, $i = 1, 2, \dots, n$.

Then define $f_\alpha(t_1, t_2, \dots, t_n) := v(\alpha)$.

The proof that f_α is well-defined, uses Proposition we proved earlier. Indeed, suppose that $v' : PV \longrightarrow \{T, F\}$ is also such that $v'(p_i) = t_i$, $i = 1, 2, \dots, n$. But then by the proposition, $v(\alpha) = v'(\alpha)$.

Notice that $v(\alpha)$ for the valuation v corresponding to the n -tuple $(t_1, t_2, \dots, t_n)$ just gives the truth value of α under the assignments t_i to its propositional variables p_i . In other words, f_α simply represents the truth table of α .

Moreover, clearly, such an f_α would be unique. □

Proposition 0.2. Given any truth function $f : \{T, F\}^n \longrightarrow \{T, F\}$, there is a wff A_f with n propositional variables such that $f_{A_f} = f$. So f represents the truth table of A_f .

Proof. Consider propositional variables $p_1, p_2, \dots, p_n$, corresponding to $i = 1, \dots, n$. If $f(t_1, t_2, \dots, t_n) = F$ for all n -tuples $(t_1, t_2, \dots, t_n)$, take $A_f := p_1 \wedge \neg p_1 \wedge p_2 \wedge p_3 \wedge \dots \wedge p_n$, for instance, so that A_f is a contradiction. It is clear in this case, that $f_{A_f} = f$. So let us assume that $f(t_1, t_2, \dots, t_n) = T$, for some $(t_1, t_2, \dots, t_n)$.

We shall construct the required wff A_f as a disjunction of conjuncts involving the p_i 's or their negations, in a manner outlined below. (We shall see later that A_f is in a very special form, called the *canonical disjunctive normal form* (CDNF)).

Observe that f may be represented like a truth table: the first n entries in any row give the values of $t_1, \dots, t_n$, and the last $(n + 1)$ th entry in that row, gives the value $f(t_1, t_2, \dots, t_n)$. There are 2^n rows in this table of f .

1. For each row, we first construct a conjunction:
the conjunct $C_i := l_1 \wedge \dots \wedge l_n$ for the i th row, $1 \leq i \leq 2^n$,
is defined such that
 $l_j := p_j$ if t_j is T and
 $l_j := \neg p_j$ otherwise,
where $j = 1, \dots, n$ and $t_1, t_2, \dots, t_n$ are the first n entries of
the row.
2. Consider all the rows for which the corresponding tuples
 $(t_1, t_2, \dots, t_n)$ are such that $f(t_1, t_2, \dots, t_n) = T$. (We have
assumed that there is at least one such row). Suppose the
conjuncts for these rows constructed as in (1) above are given
by $C_{i_1}, \dots, C_{i_k}$. Define $A_f := C_{i_1} \vee \dots \vee C_{i_k}$.

Note that

- (i) $v(C_j) = T$, for a valuation v corresponding to the j th
row, i.e. where $v(p_i) = t_i$, $i = 1, 2, \dots, n$, $t_1, t_2, \dots, t_n$ being
the first n entries of the row.
- (ii) Moreover, $v(C_k) = F$, for any $k \neq j$, and $v'(C_j) = F$,
for every v' different from v , i.e. v' such that $v'(p_i) \neq t_i$ for
some $i = 1, \dots, n$.

Now we need to prove that $f_{A_f} = f$.

For any $(t_1, t_2, \dots, t_n) \in \{T, F\}^n$, take v such that $v(p_i) = t_i$, $i = 1, 2, \dots, n$.

Then $f_{A_f}(t_1, t_2, \dots, t_n) = v(A_f)$, as given in the construction of Proposition 0.1.

Consider the i th row in the table of f , its first n entries $t_1, t_2, \dots, t_n$, and a valuation v for $(t_1, t_2, \dots, t_n)$ just defined. There are two possible cases.

Case 1: $f(t_1, t_2, \dots, t_n) = T$.

By the construction in (1), C_i occurs in A_f . We must show that $v(A_f) = T$.

As observed in (i) above, $v(C_i) = T$. As A_f is a disjunction, $v(A_f) = T$.

Case 2: $f(t_1, t_2, \dots, t_n) = F$.

C_i does not occur in A_f , and $v(C_j) = F$ for all $j \neq i$ (cf. (ii)). So $v(A_f) = F$. $\square$

For any wff α with propositional variables $p_1, \dots, p_n$, we get the truth function f_α by Proposition 0.1. Then by Proposition 0.2, we get the wff A_{f_α} for the truth function f_α , such that $f_{A_{f_\alpha}} = f_\alpha$. In other words, the truth tables of the wffs α and A_{f_α} are identical. Thus, we have the following.

Corollary 0.3. For any wff $\alpha(p_1, \dots, p_n)$, $A_{f_\alpha} \equiv \alpha$, i.e. the two wffs are logically equivalent.

Note that there is nothing sacrosanct about taking the particular propositional variables $p_1, \dots, p_n$ – the above would all hold for α with *any* n propositional variables, at best some re-labelling may have to be done. Secondly, if the set of propositional variables occurring in α happens to be a *proper* subset of $p_1, \dots, p_n$, Propositions 0.1 and 0.2 would still guarantee the existence of a truth function f such that $A_f \equiv \alpha$ and A_f contains the propositional variables $p_1, \dots, p_n$.

Example 0.1. Suppose $n = 3$, and $\alpha := p_1 \wedge p_2$. Then f and A_f are as follows.

t_1	t_2	t_3	$f(t_1, t_2, t_3)$
T	T	T	T
T	T	F	T
T	F	T	F
T	F	F	F
F	T	T	F
F	T	F	F
F	F	T	F
F	F	F	F

$$A_f := (p_1 \wedge p_2 \wedge p_3) \vee (p_1 \wedge p_2 \wedge \neg p_3).$$